h2 Estimate vs. Sample Size at Fixed, Low F_ST, by LDSC Variant

Real chr1 LD structure, F_ST=0.01, sigma_s=0.3, n=500 through n=10000 (m=2n), all six LDSC estimator variants, one section per scenario

Executive Summary

The companion reports fix sample size per figure and sweep F_ST from 0 to 0.1 (realistic_LD_sharedf_vs_indepf_summary.pdf), or fix F_ST=0.1 and compare all six LDSC estimator variants (ldsc_estimator_variants_comparison.pdf). This report fixes F_ST=0.01 (a small but nonzero stratification level) and sigma_s=0.3, and sweeps sample size n from 500 to 10000 (m=2n) on real chr1 LD structure, comparing all six LDSC estimator variants within each of the four scenarios (reference panel present or not, stratification vector f shared with the reference panel or drawn independently).

Headline finding: the six variants split into three groups with distinct n-scaling behavior, and this split holds across all four scenarios. The two **fixed-intercept** variants climb steadily and roughly monotonically with n in every scenario, from ~0.33-0.39 at n=500 to ~0.53-0.56 at n=10000 - essentially unaffected by reference-panel choice or f-sharing at this F_ST. The two **free-intercept/two-step** variants (including the project's default, h2_ldsc_reg) stay close to flat near h2_pop through n=2000 in every scenario, then rise more modestly to ~0.37-0.40 by n=10000. The two **free-intercept/single-step** variants are the most scenario-dependent: in three of the four scenarios they stay near baseline through n=5000 and then spike sharply at n=10000, reaching ~0.51-0.70 (higher than even the fixed-intercept variants in the weighted case) - but in the one scenario with *both* a reference panel and an independently-drawn f, they instead stay flat near h2_pop (~0.28-0.30) all the way to n=10000, with no late blow-up at all.

This is a different angle on the same mechanism the companion report identifies: there, the ref-panel/independent-f scenario is the one where free-intercept/single-step estimation *collapses toward zero* as F_ST grows at fixed n. Here, at a fixed, low F_ST and growing n instead, that same scenario's free-intercept/single-step estimates stay anchored near h2_pop while the *same estimators* inflate sharply with scale in every other scenario. The intercept-stabilizing mechanism that causes collapse at high F_ST appears to also suppress this n-driven blow-up at low F_ST - the ref-panel/independent-f scenario is consistently the outlier for these two estimators, just in opposite directions depending on which parameter is being varied.

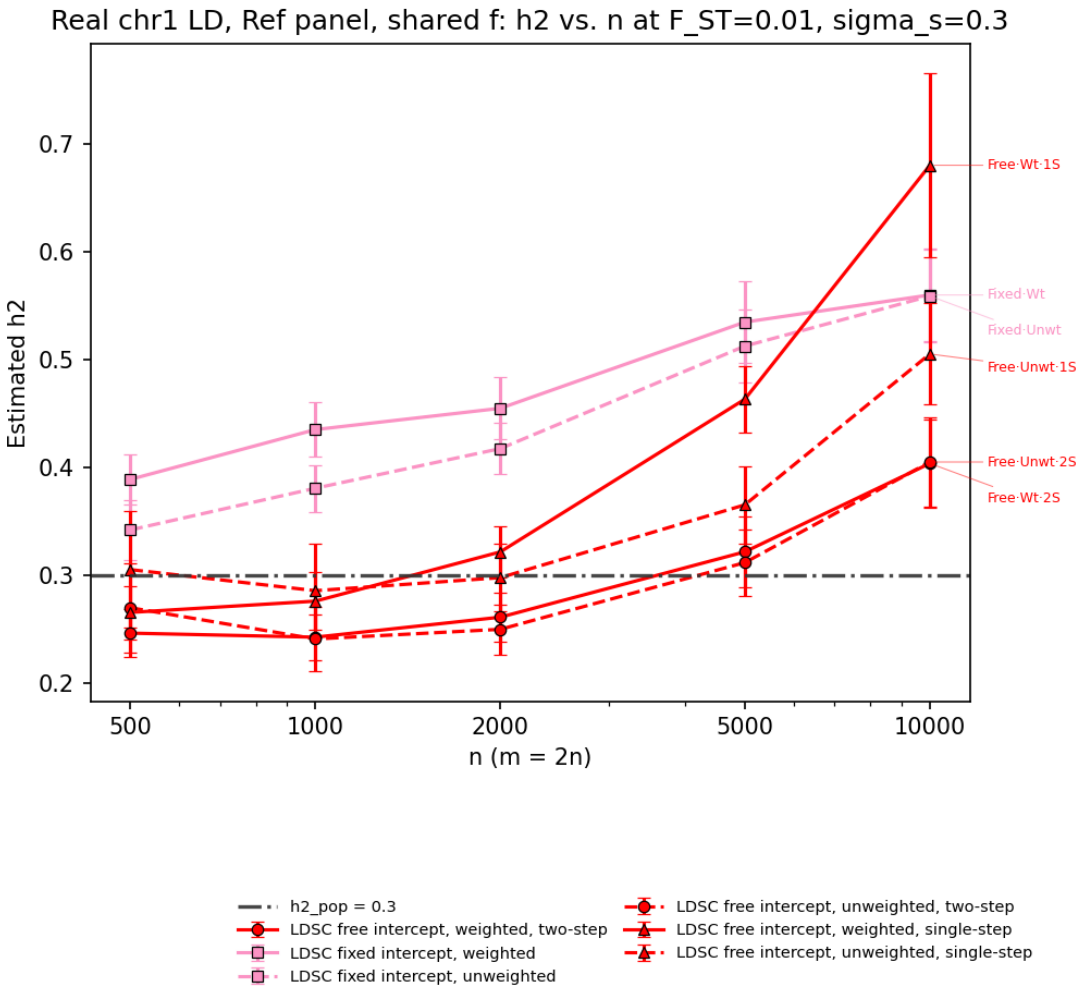
Methods

See the companion reports for full simulation methodology and the six LDSC variant definitions. Each point here is the same F_ST=0.01 dataset already saved for the main report's low-F_ST table, re-sliced to hold F_ST fixed and vary n instead of holding n fixed and varying F_ST. No new simulations were run for this report. chr22 is not included - it has not been run at n=10000.

Scenario: Ref panel, shared f

Mean h2 estimate over 100 replicates at F_ST=0.01, by sample size n and LDSC variant. h2_pop=0.3.

n	Free/Wt/2step	Fixed/Wt	Fixed/Unwt	Free/Unwt/2step	Free/Wt/1step	Free/Unwt/1step
500	0.246	0.389	0.342	0.270	0.266	0.305
1000	0.243	0.435	0.381	0.241	0.276	0.286
2000	0.261	0.455	0.418	0.250	0.322	0.298
5000	0.322	0.535	0.513	0.312	0.463	0.365
10000	0.404	0.560	0.559	0.405	0.680	0.505

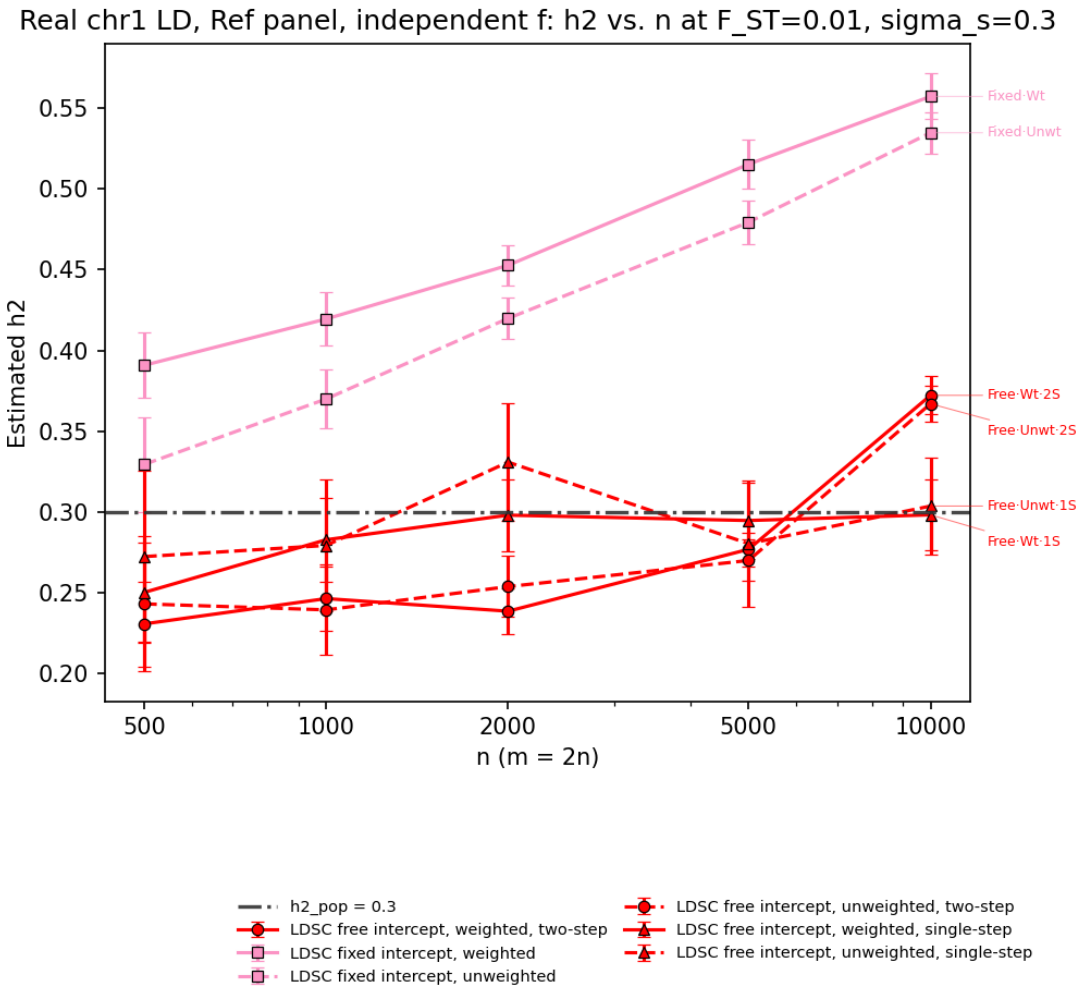


chr1, F_ST=0.01, sigma_s=0.3 — Ref panel, shared f

Scenario: Ref panel, independent f

Mean h2 estimate over 100 replicates at F_ST=0.01, by sample size n and LDSC variant. h2_pop=0.3.

n	Free/Wt/2step	Fixed/Wt	Fixed/Unwt	Free/Unwt/2step	Free/Wt/1step	Free/Unwt/1step
500	0.231	0.391	0.329	0.243	0.250	0.272
1000	0.246	0.419	0.370	0.239	0.283	0.279
2000	0.239	0.453	0.420	0.254	0.298	0.331
5000	0.277	0.515	0.479	0.270	0.295	0.280
10000	0.372	0.557	0.535	0.367	0.298	0.304

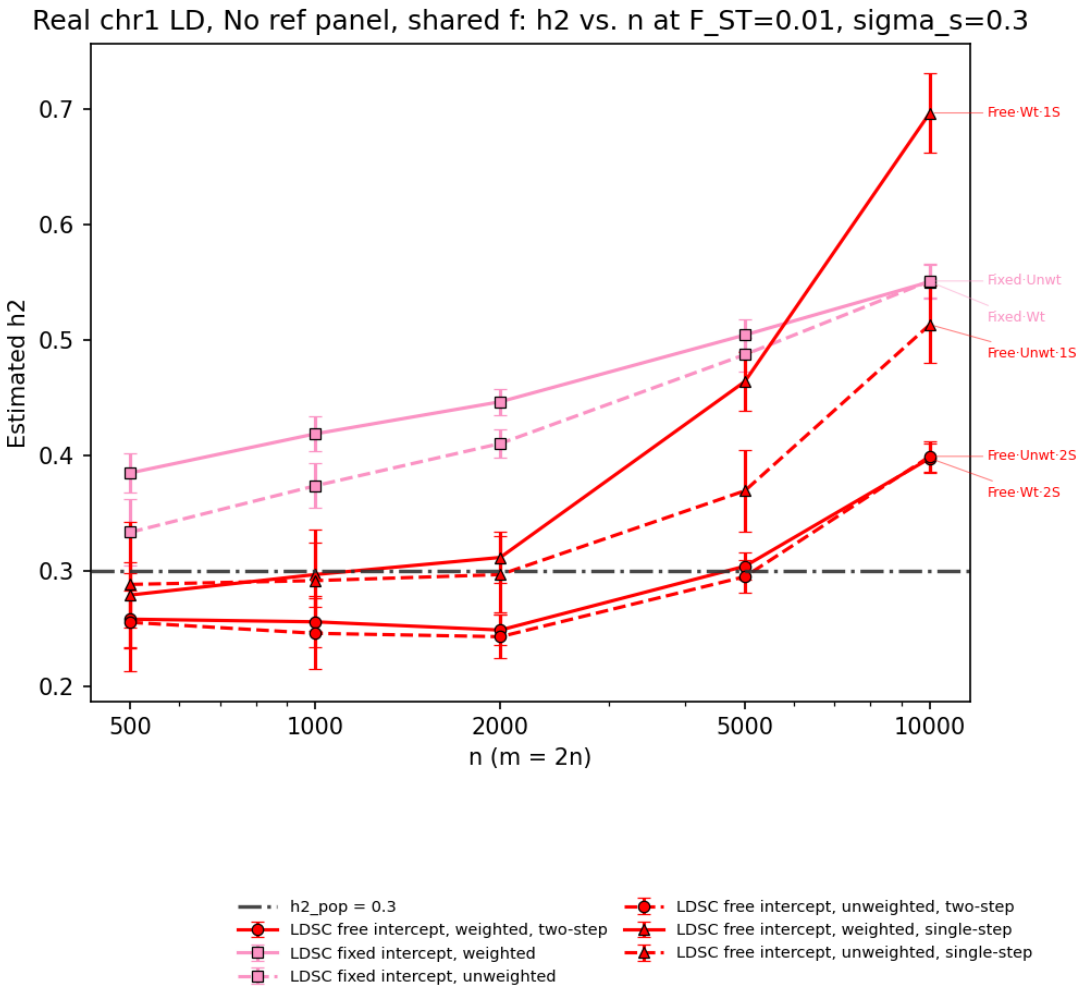


chr1, F_ST=0.01, sigma_s=0.3 — Ref panel, independent f

Scenario: No ref panel, shared f

Mean h2 estimate over 100 replicates at F_ST=0.01, by sample size n and LDSC variant. h2_pop=0.3.

n	Free/Wt/2step	Fixed/Wt	Fixed/Unwt	Free/Unwt/2step	Free/Wt/1step	Free/Unwt/1step
500	0.259	0.385	0.334	0.256	0.280	0.289
1000	0.256	0.419	0.374	0.246	0.297	0.292
2000	0.249	0.447	0.411	0.243	0.312	0.297
5000	0.304	0.505	0.488	0.295	0.464	0.370
10000	0.398	0.551	0.551	0.400	0.697	0.514

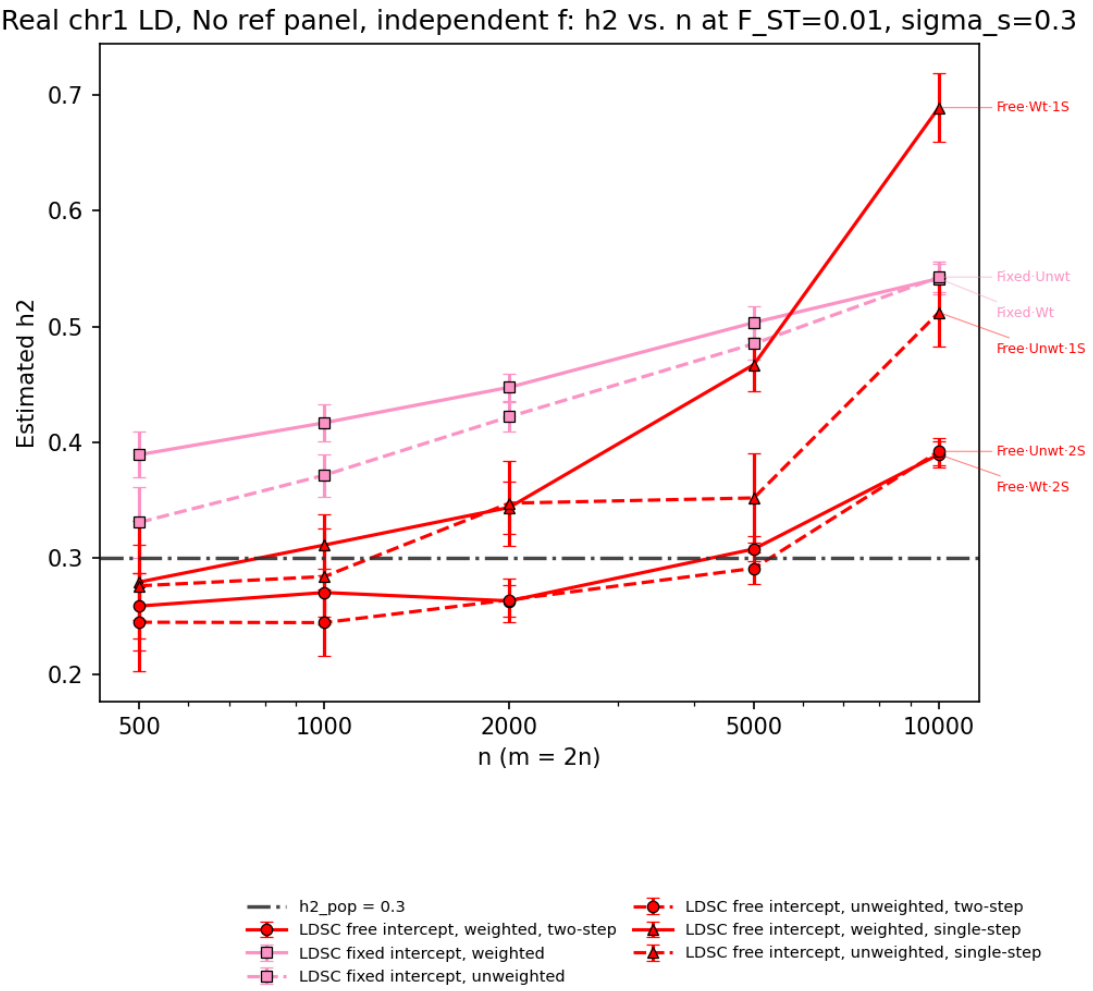


chr1, F_ST=0.01, sigma_s=0.3 — No ref panel, shared f

Scenario: No ref panel, independent f

Mean h2 estimate over 100 replicates at F_ST=0.01, by sample size n and LDSC variant. h2_pop=0.3.

n	Free/Wt/2step	Fixed/Wt	Fixed/Unwt	Free/Unwt/2step	Free/Wt/1step	Free/Unwt/1step
500	0.259	0.389	0.331	0.245	0.279	0.276
1000	0.270	0.417	0.372	0.244	0.311	0.284
2000	0.263	0.448	0.422	0.264	0.344	0.348
5000	0.308	0.504	0.485	0.291	0.467	0.352
10000	0.390	0.542	0.543	0.392	0.689	0.512



chr1, F_ST=0.01, sigma_s=0.3 — No ref panel, independent f

Discussion

Fixed-intercept variants inflate with scale regardless of scenario: both fixed-intercept variants roughly double from n=500 to n=10000 in every one of the four scenarios, including the two with no reference panel at all. Since a fixed intercept removes the regression's main degree of freedom for absorbing an f-mismatch signal, this scale-driven inflation cannot be an f-mismatch effect - it is a property of fitting LDSC with a fixed intercept to a larger sample at this F_ST, independent of reference-panel structure.

Free-intercept/two-step variants are the most stable: these track closest to h2_pop at small-to-moderate n across all four scenarios, consistent with the companion report's finding that the two-step procedure is one of the two ways (along with a fixed intercept) to stabilize the intercept and avoid the worst scale/F_ST-driven distortions.

Free-intercept/single-step variants reveal the sharpest scenario split: these are the two variants with no safeguard on the intercept at all. In three scenarios they spike to the highest h2 estimates of all six variants by n=10000 (roughly 0.51-0.70, i.e. 1.7-2.3× h2_pop); in the ref-panel/independent-f scenario specifically, they instead stay essentially flat near h2_pop through n=10000. This is the same pair of estimators the companion report shows collapsing toward zero as F_ST grows at fixed n in that same scenario - the two findings together suggest the ref-panel/independent-f mismatch specifically interferes with whatever mechanism drives these two estimators' values upward with scale elsewhere, rather than simply adding a separate bias on top of it.

Practical implication: a free intercept estimated jointly with the slope in a single pass is the least scale-robust choice among the six variants tested, by a wide margin, in every scenario except the one where a mismatched reference panel happens to suppress that instability. Relying on that apparent stability in the mismatched-reference-panel case would be a mistake - it is a side effect of a known failure mode (the collapse documented in the companion report), not evidence that the estimator is well-behaved there.